Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.
Q = Φ/(ρ c_p ΔT) when a heat load is given; v = Q/(πD²/4); Re = ρvD/μ; f = 64/Re (Re < 2300) else Colebrook–White 1/√f = −2 log10(ε/3.7D + 2.51/(Re√f)) by fixed-point iteration (≤ 100 steps) with the Swamee–Jain explicit value as start and cross-check; h_f = f (L + L_eq)/D · v²/2g + K v²/2g; H_sys(Q) = Σdz + h_out + Σh_f(Q) with h_f ∝ Q² scaled from the design flow (closed circuit: Σdz = 0, the rise in the flow is recovered in the return); the water held in the pipes V_pipe = Σ π/4·D²·L; duty point where H_pump(Q) = H_sys(Q) (bisection); affinity: Q ∝ n D³, H ∝ n² D²; NPSH_a = p_atm/ρg + z_s − h_f,suction − p_v/ρg.
Ref: Crane Technical Paper 410 — Flow of fluids through valves, fittings and pipe
Ref: Çengel & Cimbala, Fluid Mechanics Ch. 8 — Internal flow: friction factor, minor losses, pump duty
Ref: CIBSE Guide C Section 4 — Flow of fluids in pipes and ducts (principles)
Ref: Kell, J. Chem. Eng. Data 20(1), 1975 — Density of liquid water

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	custom (□ and □ given)		spec: Oil datasheet ISO VG 680 at 20 °C	any
T	Fluid temperature (mean; link B6 t_m for a heating loop)	20	degC	assumption: Ambient 20 °C.	0 ... 100 degC
rho_custom	Density (custom fluid)	888	kg/m^3	spec: Oil datasheet ISO VG 680	500 ... 2000 kg/m^3
mu_custom	Dynamic viscosity (custom fluid)	800	mPa*s	spec: Oil datasheet ISO VG 680 at 20 °C	0.1 ... 1000 mPa*s
Q	Design flow rate (blank when a heat load gives it)	1	m^3/h	spec: Transfer pump duty	0.01 ... 100000 m^3/h
segments	Pipe segments in series	1 row(s)		drawing: DRG-L-020 rev A	any
segments[1]		name = transfer line; L = 40 m; D = 50 mm; eps = 0 mm; K = 0; L_eq = 0 m; dz = 0 m; side = discharge			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
circuit	Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)	open (the pump lifts the water by the rises entered)		default	any
h_out	Pressure head required at the outlet	0	m	assumption: Open discharge.	0 … 1000 m
v_max	Velocity limit (warning)	3	m/s	assumption: Practice limit.	0.5 … 10 m/s
n_ratio	Speed ratio n_2/n_1 (affinity)	1		default	0.2 … 2
D_ratio	Impeller diameter ratio D_2/D_1 (affinity)	1		default	0.7 … 1.1
eta_pump	Pump efficiency at duty (for shaft power)	0.7		default	0.2 … 0.95
p_atm	Atmospheric pressure (NPSH)	101.325	kPa	default	50 … 110 kPa
z_s	Static head at the pump suction (+ flooded, − lift)	0	m	default	-10 … 100 m

3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales friction losses with Q^2 from the design flow (turbulent assumption); the static part is independent of flow. ($H_{sys} = H_{static} + k\,Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. T (Fluid temperature (mean; link B6 t_m for a heating loop)) is taken as 20 degC. (Ambient 20 °C.)
A6. h_out (Pressure head required at the outlet) is taken as 0 m. (Open discharge.)
A7. v_max (Velocity limit (warning)) is taken as 3 m/s. (Practice limit.)

4. Working

[5] rho = rho_custom
= 888 kg/m³
= 888 kg/m³
Density (as entered)
[6] mu = mu_custom
= 800 mPa·s
= 800 mPa·s
Dynamic viscosity (as entered)
[10] Q_design = Q
= 1 m³/h
= 1 m³/h
Design flow (as entered)

[11] v_s1 = transfer line: Q / (π D²/4)
= 1 m³/h / (π × 50² mm² / 4)
= 0.14147 m/s
Velocity, transfer line
[12] Re_s1 = transfer line: ρ v D / μ
= 888 × 0.14147 × 0.05 / 0.8
= 7.8516
Reynolds number, transfer line
[13] f_s1 = transfer line: Darcy friction factor
= 64 / Re (laminar)
= 8.1512
Friction factor, transfer line
[14] h_f_s1 = transfer line: f (L + L_eq)/D · v²/2g + K v²/2g
= 8.1512 × (40 + 0) / 0.05 × 0.0010204 + 0 × 0.0010204
= 6.6542 m
Head loss, transfer line
[15] h_f_total = Σ h_f
= 1 segment(s)
= 6.6542 m
Total friction and fitting loss at the design flow
[17] V_pipe = Σ π/4·D²·L
= 1 segment(s)
= 78.54 L
Water held in the pipe segments
[18] H_static = Σ dz + h_out
= 0 + 0
= 0 m
Static head
[19] H_sys = H_static + h_f_total
= 0 m + 6.6542 m
= 6.6542 m
System head at the design flow
[20] dp_total = rho * 9.80665 m/s^2 * h_f_total
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
Friction pressure drop at the design flow
[21] dp_sys = rho * 9.80665 m/s^2 * H_sys
= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m
= 57.947 kPa
System pressure at the design flow
[22] P_hyd = rho * 9.80665 m/s^2 * Q_design * H_sys
= 888 kg/m³ × 9.80665 m/s^2 × 1 m³/h × 6.6542 m
= 0.016096 kW
Hydraulic power at the design point
[25] v_peak = highest segment velocity
= 0.14147 m/s
= 0.14147 m/s
Highest velocity
[26] u_v = v_peak / v_max
= 0.14147 / 3
= 0.047157
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
v_s1	0.14147 m/s	Velocity, transfer line
Re_s1	7.8516	Re, transfer line
f_s1	8.1512	f, transfer line
h_f_s1	6.6542 m	h_f, transfer line
h_f_total	6.6542 m	Σ h_f
H_static	0 m	Static head
V_pipe	78.54 L	Water held in the pipes
H_sys	6.6542 m	System head at Q
dp_total	57.947 kPa	Friction Δp
dp_sys	57.947 kPa	System Δp
Q_design	1 m³/h	Design flow
P_hyd	0.016096 kW	Hydraulic power
v_peak	0.14147 m/s	Highest velocity

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Velocity ≤ limit	u_v = 0.047157	u_v ≤ 1	0.0472	

7. Applicability

WARNING — input.circuit: circuit (Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)) was not entered; the template default 'open (the pump lifts the water by the rises entered)' is used.
WARNING — input.n_ratio: n_ratio (Speed ratio n_2/n_1 (affinity)) was not entered; the template default 1 is used.
WARNING — input.D_ratio: D_ratio (Impeller diameter ratio D_2/D_1 (affinity)) was not entered; the template default 1 is used.
WARNING — input.eta_pump: eta_pump (Pump efficiency at duty (for shaft power)) was not entered; the template default 0.7 is used.
WARNING — input.p_atm: p_atm (Atmospheric pressure (NPSH)) was not entered; the template default 101.3 kPa is used.
WARNING — input.z_s: z_s (Static head at the pump suction (+ flooded, − lift)) was not entered; the template default 0 m is used.
WARNING — laminar: segments: transfer line is laminar (Re = 7.8516); f = 64/Re.

8. Verification

Template bs.pipe-pump@1.3.0 Content hash d3c3c38f4c81cf2d Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0